

## BRST Symmetry:

Start with original Polyv. Action

$$S^{\text{orig.}} = -\frac{1}{4\alpha'} \int d^2\sigma \sqrt{-g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}$$

• This action has the following symmetries

(i) Reparametrization:

$$\delta g_{\alpha\beta} = \nabla_\alpha V_\beta + \nabla_\beta V_\alpha$$

$$\delta X^\mu = V^\alpha \partial_\alpha X^\mu$$

→  $\mu$ : is invariant under a general coord transf.

$$g^{\alpha\beta} = -(\nabla^\alpha V^\beta + \nabla^\beta V^\alpha)$$

$$\delta g = g g^{\rho\sigma} \delta g_{\rho\sigma} = 2g \nabla_\alpha V^\alpha$$

(ii) Weyl invariance:

$$\begin{cases} \delta g_{\alpha\beta} = c^\phi g_{\alpha\beta} \\ \delta X^\mu = 0 \end{cases}$$

\* We now Gauge-fix the above action by introducing Ghost fields:  $(b, c)$ . This will leave the prop. invariant.

→

## BRST PRIMER

$$A^x B_x \equiv \int dx A(x) B(x)$$

• Gauge symmetry:  $\delta \Phi^F = \epsilon^\alpha \delta_\alpha \Phi^F$   
 $\epsilon^\alpha = \epsilon^\alpha(x)$

• Invariance of this Gauge symmetry

$$S[\Phi^F + \delta \Phi^F] = S[\Phi^F] \Rightarrow \delta S[\Phi^F] = 0$$

• Gauge-fixing condition:

$$F^A(\Phi^F) = 0$$

Rules:

### (i) Introduce Ghost fields:

$$e^{\alpha} \rightarrow c^{\alpha} \quad \forall \alpha$$

$c^{\alpha}$  has opposite statistics to  $e^{\alpha}$

ex: If  $e^{\alpha} = (\lambda^1, \lambda^2)$   $\lambda^{\alpha} \lambda^{\beta} = \lambda^{\beta} \lambda^{\alpha}$

Then  $c^{\alpha} = (c^1, c^2)$   $c^{\alpha} c^{\beta} = -c^{\beta} c^{\alpha}$

### (ii) Introduce Anti-Ghost fields:

- $b_{\alpha}$  as well as Auxiliary fields  $B_A$ , one for each Gauge fixing condition

$$F^A(\Phi^{\pm}) = 0$$

$$\begin{aligned} b_A &\leftrightarrow c_A \\ B_A &\leftrightarrow G_A \end{aligned}$$

 $\leftarrow \begin{cases} \text{Same} \\ \text{Statistics} \end{cases}$

### (iii) We add the following terms to $S^{\text{orig}}$ :

$$\begin{aligned} (a) \quad S^{\text{gf}} &= -i B_A F^A(\Phi^{\pm}) \\ (b) \quad S^{\text{gh}} &= b_A c^{\alpha} \partial_{\alpha} F^A \end{aligned}$$

[Gauge Fixing]  
[Ghost Action]

$$S = S^{\text{or}} + S^{\text{gh}} + S^{\text{gf}}$$

### Back to the String:

\* Recall that we may locally find a  $f^{\mu} \phi(\xi)$   
 $\phi: \Sigma \rightarrow \mathbb{R}$ , s.t.  $g_{\alpha\beta}$  is locally conformally equivalent to  $\eta_{\alpha\beta}$

$$g_{\alpha\beta} = e^{\phi(\xi)} \eta_{\alpha\beta}$$

→ To rid ourselves of  $\phi$  is our choice we look to Gauge fix the Weyl invariant term

$$\sqrt{-g} g^{\alpha\beta} = \sqrt{-\hat{g}} \hat{g}^{\alpha\beta}$$

$$A = (\alpha\beta)$$

$$F^A(g) = 0 \equiv \sqrt{-\hat{g}} \hat{g}^{\alpha\beta} - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta} = 0$$

(Gauge Fixing Condition)

$$F_A(g) = F_{AB}(g) = 0$$

- From the BRST procedure this implies we introduce the action for Gauge fixing

$$S^{gf} = -i B_{AB} F^{AB}(\hat{g}, \eta) \\ = -i \int d^2 \xi B_{AB} (\sqrt{-g} g^{AB} - \sqrt{-\hat{g}} \hat{g}^{AB})$$

→ We normalize this as

$$S^{gf} = -\frac{i}{\pi} \int d^2 \xi \lambda_{AB} (\sqrt{-g} g^{AB} - \sqrt{-\hat{g}} \hat{g}^{AB})$$

Where  $B_{AB} i = \lambda_{AB} / \pi$

$\lambda_{AB}$ : (Auxiliary) Lagrange field

\* The Constraint  $F_{AB} = 0$  contains only two conditions since we are choosing

$$\sqrt{-g} g_{AB} = \eta_{AB} \quad \text{and so}$$

$$\sqrt{-g} \begin{pmatrix} g_{00} & 0 \\ 0 & g_{11} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{cases} \sqrt{-g} g_{00} = -1 \\ \sqrt{-g} g_{11} = 1 \end{cases}$$

- Only two components of  $\lambda_{AB}$  contribute to the Action.

→  $\lambda_{AB} = \lambda_{BA}$  and  $\text{tr}(\lambda_{AB}) = 0$  with respect to either  $g^{AB}$  or  $\hat{g}^{AB}$ .

\* For Diffeomorphism invariance:

$$\delta x^\mu = V^\alpha \partial_\alpha x^\mu$$

$1/V^\alpha \leftrightarrow F^\alpha$  This is the numerator

$$V^\alpha \mapsto \Lambda C^\alpha$$

Thus,

$$\delta X^\mu = \Lambda C^\alpha \partial_\alpha X^\mu$$

$$\delta g_{\alpha\beta} = \nabla_\alpha (\Lambda C_\beta) + \nabla_\beta (\Lambda C_\alpha)$$

$$\rightarrow \delta X^\mu = \Lambda C^\alpha \partial_\alpha X^\mu$$

$$\delta g^{\alpha\beta} = -\Lambda (\nabla^\alpha C^\beta + \nabla^\beta C^\alpha)$$

\* The variation of all the fields

$$\delta_B b_{\alpha\beta} = \Lambda \lambda_{\alpha\beta}$$

$$\delta_B \lambda_{\alpha\beta} = 0$$

$$\delta_B C^a = -\frac{1}{2} \kappa f_{\gamma\delta}^a C^\gamma C^\delta$$

From these

$$\delta_B b_{\alpha\beta} = \Lambda \lambda_{\alpha\beta}$$

$$\delta_B \lambda_{\alpha\beta} = 0$$

(other ghost fields)  
BRST

• This implies:  $b_{\alpha\beta} = -b_{\beta\alpha}$ ,  $\text{tr}(b) = 0$

$$\text{We take: } \delta_B C^a = \Lambda C^\beta \nabla_\beta C^a = \Lambda C^\beta \partial_\beta C^a$$

$$[\delta_1, \delta_2] X^\mu = [E_1^\alpha \partial_\alpha, E_2^\beta \partial_\beta] X^\mu$$

$$= E_1^\alpha \partial_\alpha E_2^\beta - (1 \leftrightarrow 2)$$

$$= (E_1^\alpha \nabla_\alpha E_2^\beta - (1 \leftrightarrow 2)) X^\mu$$

$$\bullet E_1^\alpha \rightarrow \Lambda C^\alpha$$

$$E_2^\beta \rightarrow C^\beta$$

$$[\delta_1, \delta_2] X^\mu = \Lambda C^\alpha \nabla_\alpha C^\beta - C^\alpha \nabla_\alpha (\Lambda C^\beta)$$

$$= (2 \Lambda C^\alpha \nabla_\alpha C^\beta)$$

• Fields  $\delta C^a$

\* Note: BRST is nilpotent:

$$\begin{aligned}\delta_1 \delta_2 X^\mu &= \delta_1 (\Lambda_2 \partial^a \partial_a X^\mu) \\ &= \Lambda_2 \delta_1 \partial^a \partial_a X^\mu + \Lambda_2 \partial^a \partial_a (\delta_1 X^\mu) \\ &= \Lambda_2 (\Lambda_1 \partial^\gamma \partial_\gamma \partial^a \partial_a X^\mu) \\ &\quad + \Lambda_2 \partial^a \partial_a (\Lambda_1 \partial^a \partial_a X^\mu)\end{aligned}$$

Now add the Ghost Action:

$$\begin{aligned}S^{\text{gh}} &= b_{\alpha\beta} \overbrace{C^\gamma \partial_\gamma F^{\alpha\beta}} \quad \leftarrow \text{Int. over } \int d^2\tau \\ &= b_{\alpha\beta} C^\gamma \partial_\gamma (\sqrt{-g} g^{\alpha\beta} - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta})\end{aligned}$$

$$S^{\text{gh}} = \int d^2\tau \underbrace{b_{\alpha\beta} C^\gamma \partial_\gamma (\sqrt{-g} g^{\alpha\beta} - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta})}_{\text{Int. over } \int d^2\tau}$$

$$\begin{aligned}\text{i)} \quad \partial_\gamma (\sqrt{-g}) &= -\frac{1}{2} \frac{1}{\sqrt{-g}} \partial_\gamma g = -\frac{1}{\sqrt{-g}} g^{\alpha\beta} \partial_\gamma g_{\alpha\beta} \\ \partial_\gamma (\sqrt{-g}) &= \sqrt{-g} \cdot \Lambda \nabla_\gamma C^\alpha\end{aligned}$$

$$\begin{aligned}\text{ii)} \quad \partial_\gamma g^{\alpha\beta} &= -(\nabla^\alpha (\Lambda C^\beta) + \nabla^\beta (\Lambda C^\alpha)) \\ &= -\Lambda (\nabla^\alpha C^\beta + \nabla^\beta C^\alpha)\end{aligned}$$

\* Derivation of Ghost Action:

• We have

$$S_3 = b_{\alpha\beta} \overbrace{C^\gamma \partial_\gamma F^{\alpha\beta}} \quad (\text{FF})$$

Note: For EACH index summation we integrate over different coords

$$\begin{aligned}b_{\alpha\beta} C^\gamma \partial_\gamma F^{\alpha\beta} &= \int d^2\tau' \int d^2\tau \underbrace{b_{\alpha\beta}(\tau') C^\gamma(\tau) \partial_\gamma(\tau) F^{\alpha\beta}(\tau')} \\ &= \int d^2\tau \int d^2\tau' b_{\alpha\beta}(\tau') C^\gamma(\tau) \partial_\gamma(\tau) (\sqrt{-g} g^{\alpha\beta}(\tau') - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta}(\tau')) \\ &= \int d^2\tau \int d^2\tau' b_{\alpha\beta} C^\gamma \left[ -\frac{1}{2} \frac{1}{\sqrt{-g}} \partial_\gamma g g^{\alpha\beta} + \sqrt{-g} \partial_\gamma g^{\alpha\beta} \right]\end{aligned}$$

\* Must understand  $\partial_\gamma g$  etc...

Note  
 $\delta g^{\alpha\beta} = -1(\nabla^\alpha V^\beta + \nabla^\beta V^\alpha)$   
 but  $\delta g^{\alpha\beta} = 4\pi \delta_\rho g^{\alpha\beta}$ , so using

$$\delta g^{\alpha\beta} = -(\nabla^\alpha V^\beta + \nabla^\beta V^\alpha) = \int d^2\xi' V^\rho(\xi) \delta_\rho(\xi) g^{\alpha\beta}$$

It's easy to see LHS:

$$-\int d^2\xi' (\nabla^\alpha V^\beta + \nabla^\beta V^\alpha)(\xi') \delta^2(\xi - \xi')$$

$$= \int d^2\xi' (V^\beta \nabla^\alpha \delta^2(\cdot) + V^\alpha \nabla^\beta \delta^2(\cdot))$$

$$= \int d^2\xi' (\delta_\rho^\beta V^\rho \nabla^\alpha \delta^2 + \delta_\rho^\alpha V^\rho \nabla^\beta \delta^2)$$

$$= \int d^2\xi V^\rho (\delta_\rho^\beta \nabla^\alpha \delta^2 + \delta_\rho^\alpha \nabla^\beta \delta^2)$$

We conclude

$$(\delta_\rho g^{\alpha\beta})(\xi) = \delta_\rho^\beta \nabla^\alpha \delta^2(\xi - \xi') + \delta_\rho^\alpha \nabla^\beta \delta^2(\xi - \xi')$$

Since

$$\delta g^{\alpha\beta} = -g^{\alpha\gamma} \delta g_{\gamma\delta} g^{\delta\beta} \text{ we also have}$$

(as in the set of notes)

$$-g^{\alpha\gamma} \delta g_{\gamma\delta} g^{\delta\beta} = [\delta_\rho^\beta (\nabla^\alpha \delta^2) + \delta_\rho^\alpha (\nabla^\beta \delta^2)]$$

$$(\delta_{\hat{\alpha}} \delta_\rho g_{\gamma\delta} \delta_{\hat{\beta}}^\beta) = -(g_{\alpha\hat{\alpha}} g_{\beta\hat{\beta}}) [ \quad ]$$

$$\delta_\rho g_{\hat{\alpha}\hat{\beta}} = -[g_{\hat{\beta}\rho} (\nabla_{\hat{\alpha}} \delta^2) + g_{\rho\hat{\alpha}} (\nabla_{\hat{\beta}} \delta^2)]$$

$\hat{\alpha} \rightarrow \alpha, \hat{\beta} \rightarrow \beta$ : we conclude:

$$\delta_\rho g_{\alpha\beta}(\xi) = -g_{\alpha\rho}(\xi) (\nabla_\beta \delta^2(\xi - \xi')) - g_{\beta\rho} \nabla_\alpha \delta^2(\xi - \xi')$$

\* If we now gather all of the above we find that:

$$b_A c^\lambda \delta_\lambda F^A = \int d^2 \xi d^2 \xi' b_{\alpha\beta}(\xi) c^\lambda(\xi') \delta_\lambda(\xi') F^{\alpha\beta}$$

$$= \int (d^2 \xi d^2 \xi') b_{\alpha\beta}(\xi) c^\lambda(\xi') \left[ \delta_\lambda g(-\frac{1}{2}) \frac{1}{\sqrt{-g}} + \sqrt{-g} \delta g^{\alpha\beta} \right]$$

$$\rightarrow [g g^{\gamma\delta} \delta_\lambda g_{\gamma\delta} g^{\alpha\beta} (-\frac{1}{2}) \frac{1}{\sqrt{-g}} + \sqrt{-g} \delta_\lambda g^{\alpha\beta}]$$

$$= [(\sqrt{-g}/2) g^{\gamma\delta} \delta_\lambda g_{\gamma\delta} + \sqrt{-g} (-g^{\alpha\gamma} \delta_\lambda g_{\gamma\delta} g^{\beta\delta})]$$

$$= (\sqrt{-g}) (g^{\gamma\delta} g_{\gamma\delta} / 2 - g^{\alpha\gamma} g^{\beta\delta}) \delta_\lambda g_{\gamma\delta}$$

$$= \sqrt{-g} [\frac{1}{2} g^{\gamma\delta} g^{\alpha\beta} - g^{\alpha\gamma} g^{\beta\delta}] (-g_{\gamma\lambda} \nabla_\delta \delta^2 - g_{\delta\lambda} \nabla_\gamma \delta^2)$$

Plugging this back in:

$$\int d^2 \xi d^2 \xi' b_{\alpha\beta} c^\lambda (\sqrt{-g}) (\frac{1}{2} g^{\gamma\delta} g^{\alpha\beta} - g^{\alpha\gamma} g^{\beta\delta}) \left( \frac{\delta g}{\sqrt{-g}} \right)$$

- Note we required  $\text{tr}(b) = 0 = g^{\alpha\beta} b_{\alpha\beta}$  so the first term vanishes, bruh.

$$\int d^2 \xi d^2 \xi' b^{\gamma\delta}(\xi) c^\lambda(\xi') \sqrt{-g} (g_{\gamma\lambda} \nabla_\delta \delta^2 + g_{\delta\lambda} \nabla_\gamma \delta^2)$$

$$\nabla_\delta = \nabla_\delta(\xi')$$

→ The final term  $-g^{\alpha\beta} \nabla_\delta c^\delta$  is not present: ...  
Not sure where this comes from, ...

$$S^{\text{gh}} = -\frac{1}{\pi} \int d^2 \xi \sqrt{-g} b_{\alpha\beta} (\nabla^\alpha c^\beta + \nabla^\beta c^\alpha - g^{\alpha\beta} \nabla_\delta c^\delta)$$

oh...

the final term is the trace term which I got rid of.... Not sure why he's keeping it if we had  $\text{tr}(b) = 0$ .... okay.

\*Quicker Method:

$$\Lambda(Sg^h + Sg^f) = -\frac{1}{\pi} \delta \int d^2 \xi \, b_{\alpha\beta} (\sqrt{-g} g^{\alpha\beta} - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta})$$

$$= -\frac{1}{\pi} \delta_B \int d^2 \xi \, b_{\alpha\beta} F^{\alpha\beta}$$

and since  $\delta^2 = 0$

$$\delta[\Lambda(Sg^h + Sg^f)] = 0$$

$$\Lambda(c^a)^\dagger = c^a \Leftrightarrow b_{\alpha\beta}^\dagger = -b_{\alpha\beta}$$

$$\Lambda^\dagger = -\Lambda$$

## Geometric Viewpoint of BRST:

$$\int D X^M D g_{\alpha\beta} e^{i S_{\text{orig}}} = Z$$

However there are redundancies in this sum as we are counting elements which are equivalent up to Weyl/Diffeos. To account for this we integrate over a space of reps for each Gauge equivalent term

→ Restrict the integral to the Quotient Space

$$(E \times M_g) / G$$

$M_g \equiv$  (Space of Metrics on  $\Sigma_g$ )

$E \equiv$  (Space of all embedding maps  
 $X: \Sigma_g \rightarrow \mathbb{R}^D$ )

$G \equiv \text{Diff}(\Sigma_g) \times \text{Weyl}(\Sigma_g)$

• The Action of  $G$  on  $E \times M_g$  is defined

$$\text{by } f: \Sigma_g \rightarrow \Sigma_g : \gamma \rightarrow f^* \gamma$$

$$(f, e^f)(X, \gamma) \equiv (f^* X, e^f f^* \gamma)$$

\* If we now apply this to our Path mt.

$$\int \mathcal{D} X \equiv \text{Int. over } E$$

$$\int Dg_{ab} \equiv \text{Int. over } M_g$$

Thus, the correct choice of int. over eq. classes:

$$\int [Dx][Dg] = \int_{\text{ex} M_g} D^x Dg / \text{Vol}(\text{Diff}(\Sigma_g) \times \text{Weyl}(\Sigma_g))$$

\* We can now introduce Fad. P.P.:

$$1 = \Delta_{FP} \int d^2 \zeta \delta(\sqrt{-g} g_{\alpha\beta}^{\zeta} - \sqrt{\bar{g}} \bar{g}^{\alpha\beta})$$

where  $\zeta$ : Diffeomorphism

$$\zeta: g \mapsto g^{\zeta} \equiv e^{\frac{1}{2}(\partial_{\xi}^2 / \partial \xi'^2)} g(\xi)$$

$\rightarrow$  So  $\zeta \in G$ , i.e. this is the group action on  $M_g \times \Sigma$ .

## An aside on F.P. Ghosts & determinants

\* Consider  $\mathcal{H} = \mathcal{H}_{YM}[A_\mu]$

$$S[A_\mu^U] = S[A_\mu]$$

$$U(x) = \exp(-g T^a \Lambda_a(x))$$

$\rightarrow \int DU \equiv$  (Integration over the Gauge group)

(For us this is, of course,  $G = \text{Diff}_\Sigma \times \text{Weyl}$ )

\* If we int. over the edg. instead, i.e.

$$\int DU = \prod_a \int d\Lambda_a$$

\* Also, we assume invariance under

$$\int DU f(U) = \int DU f(U U')$$

i.e. it is (left) invariant under the group action.

\* Consider:

$$\Delta[A_\mu] = \int DU \delta[F_a(A_\mu^U)]$$

Functional  $\delta$ -f

$$\Delta[A_\mu] = \int DU \delta[F_a(A_\mu^U)]$$

$$\delta[F_a] = \frac{1}{a} \delta[F_a]$$

- $\Delta[A_{\mu\nu}]$  is invariant under Gauge transf.'s  
 $\Leftrightarrow \Delta[A_{\mu\nu}^0] = \Delta[A_{\mu\nu}]$

- We will have

$$\Delta^{-1}[H_{\mu\nu}] \Delta[A_{\mu\nu}] = 1$$

With this we rewrite the above as:

$$Z = \int_{A \rightarrow A^0} DA^\mu \Delta^{-1}[A_{\mu\nu}] \int DU \delta[F_a(A_{\mu\nu}^0)] e^{iS}$$

$$\begin{aligned} \int DA^\mu e^{iS[A]} &= \int DA^\mu \Delta^{-1}[A_{\mu\nu}] \int DU \delta[F_a(A_{\mu\nu}^0)] e^{iS[A^0]} \\ &= \int DU \int DA^\mu \Delta^{-1}[A_{\mu\nu}] \delta[F_a(A_{\mu\nu})] e^{iS[A]} \end{aligned}$$

The DU integral has factored out of the expression:

Thus:

$$\int DU = \int \pi D\Lambda_a$$

We would like to have this integral be over Gauge fixed terms  $\int D F_a$

$$\delta\Lambda_a(x) = \int d^4y \frac{\delta\Lambda_a(x)}{\delta F_b(y)} \delta F_b(y)$$

However for integrals we know we must have  $\left| \frac{\partial(y^\mu)}{\partial x^\mu} \right| = \text{"det. of Jacobian"}$

$$\int DU = \int \pi_a D\Lambda_a = \int \pi_a D F_a \det \left( \frac{\delta\Lambda_a(x)}{\delta F_b(x')} \right)$$

This implies

$$\Delta[A_{\mu\nu}] = \int DU \delta F = \int \pi_a D F_a \det \left( \frac{\delta\Lambda_a}{\delta F_b} \right) \delta F$$

$$\left[ \delta\Lambda_b(x) \right]$$

$$\delta = \frac{\delta}{\delta F_b}$$

$$\Delta[A_{\mu a}] = \det \left[ \frac{\delta F_a(x')}{\delta A_b(x)} \right] \Big|_{F_a=0}$$

Using  $\Delta^{-1} \Delta = 1$  we have

$$\Delta^{-1}[A_{\mu a}] = \det \left( \frac{\delta F_a(x')}{\delta A_b(x)} \right) \Big|_{F_a=0}$$

## Back to the string:

\* Looking at the above it seems rather childish. Instead let us be geometric (See Nakamura):

On the tangent space of  $\Sigma_g$ :  $T\Sigma_g$  we have the inner product

$$(\delta\phi_1, \delta\phi_2)_g := \int d^2\sigma \sqrt{g} g(\delta\phi_1, \delta\phi_2)$$

• We can normalize this via

$$\int D_g \delta\phi e^{-\frac{1}{2}(\delta\phi, \delta\phi)_g} = \frac{1}{\sqrt{\det(g)}}$$

$\Leftrightarrow$  This induces a measure on the field space  $\int D\phi \sqrt{\det(g)}$

Scalar:  $(\delta f, \delta f)_g = \int d^2\xi \sqrt{g} \delta f^2$

Vector:  $(\delta V^a, \delta V^b)_g = \int d^2\xi \sqrt{g} g_{ab} \delta V^a \delta V^b$

Symm.-tensor:

$$(\delta T_{ab}, \delta T_{ab})_g = \int d^2\xi \sqrt{g} G^{abcd} T_{ab} T_{cd}$$

where  $G^{abcd} = G_+^{abcd} + 2g^{ab}g^{cd}$

$$G_+^{abcd} = g^{ac}g^{bd} + g^{ad}g^{cb} - g^{ab}g^{cd}$$

$G^{abcd}$ : de-Witt Metric

$G_+$ : Projection onto the traceless component of a tensor

ex:  $g^{ab} T_{ab} = 0 \quad \Delta g_{ab} \Rightarrow g^{ab} g_{ab} \Delta = D \Delta$

$$\begin{cases} G^{abcd} T_{cd} = G^{abcd} T_{cd} = 2 T^{ab} \\ G^{abcd} \Delta g_{cd} = 2 \Delta g^{ab} \end{cases}$$

- The vector measure is  $\text{Weyl}(\Sigma_g)$  invariant
- The Metric has the decomposition

$$\delta g_{ab} = \underbrace{\delta \Delta g_{ab}}_{\text{trace}} + \underbrace{\delta g_{ab}^\perp}_{\text{traceless}}$$

$$g^{ab} \delta g_{ab} = 2 \delta \Delta \Rightarrow \delta \Delta = \frac{1}{2} g^{ab} \delta g_{ab}$$

$$g^{ab} \delta g_{ab} = 0$$

\* We can use the above metric for  $\delta g_{ab}$

$$\begin{aligned} |\delta g_{ab}|_g^2 &= (\delta g_{ab}, \delta g_{ab})_g \\ &= 4 \Delta |\delta \Delta|_g^2 + |\delta g_{ab}^\perp|_g^2 \end{aligned}$$

→ This turns into products/sums over scalar/traceless-symm.

\* This implies the measure factors as

$$Dg_{ab} = D\Delta Dg_{ab}^\perp$$

F.P. Gauge fixing:  $(f^*g, f^*\psi) \sim (g, \psi)$

$\Leftrightarrow$  Extract

$$d(\text{fields}) = |\text{Jacobian}| \times D(\text{Gauge}) \times D(\text{physical})$$

\* Moduli-Space: Gauge equivalent metrics

$$\mathcal{M}_g \equiv \text{Met}(\Sigma_g) / G$$

Endow  $\mathcal{M}_g$  with <sup>local</sup> coordinates  $t_i$

$t_i \equiv \text{Moduli parameters}$ ,  $i = 1, \dots, \dim(\mathcal{M}_g)$

Teichmüller Space:  $\mathcal{T}_g \equiv \text{Met}(\Sigma_g)/G_0$   
 $G_0 = \text{GL}(n, \mathbb{R})$

\* Modular group:  $\Gamma_g \equiv \pi_0(\text{Diff}(\Sigma_g)) = \frac{\text{Diff}(\Sigma)}{\text{Diff}_0(\Sigma)}$

$$\mathcal{M}_g = \mathcal{T}_g / \Gamma_g$$

$$\mathcal{M}_g \equiv \dim_{\mathbb{R}}(\mathcal{M}_g) = \dim_{\mathbb{R}}(\mathcal{T}_g)$$

Their volumes are related by

$$\int_{\mathcal{M}_g} D^{\mathcal{M}_g} t = \frac{1}{\Omega_{\mathcal{T}_g}} \int_{\mathcal{T}_g} d^{\mathcal{M}_g} t$$

Def: Let  $G$  be a group, then the Haar Measure  $(dg)$ , which is left-invariant, can be integrated over the entire group  $G$  to yield the volume:

$$\Omega_G \equiv \int_G dg = \int d(hg) \quad \forall h \in G$$

Now, given the Lie Alg. of  $G$ ,  $\mathfrak{g}$ , with generators  $T_i$  we have

$$\alpha = \alpha^i T_i \in \mathfrak{g}$$

Thus, we can parametrize the integral using the coefficients  $\alpha^i$

$$\Omega_G = \int d\alpha = \int \prod_i d\alpha^i$$

Claim:  $\mathcal{T}_g$  is a mfd while  $\mathcal{M}_g$  is NOT for  $g \geq 2$ . Here  $\mathcal{M}_g$  is an orbifold.

\* Now

$$g_{ab} = \hat{g}_{ab}(\phi(t)) = e^{2\phi} \hat{g}_{ab}(t) = f^*(e^{2\phi} \hat{g}_{ab})$$

or

$$g_{ab}(x) = \tilde{g}_{ab}^{(t,t)}(x, t)$$

- Now, we are going from

$$g \rightarrow [g]$$

The space  $M_g$  has coordinates  $t_i$  and so let us Recall:

- Take  $x^M \rightarrow x^M + \delta x^M \in M$

then  $\delta x^M$  is a tangent vector on

$TM$  which is subject to  $g: TM \times TM \rightarrow K$

Now,  $M_g$  has coordinates  $t_i$  (which represent  $[\tilde{g}_{ab}] = t_i$  equivalence classes).

- ① Thus,  $t_i \rightarrow t_i + \delta t_i$  then

$\delta t_i \in TM_g$  is a tangent space element, which is subject to a Differential metric.

- ② Note, however, what  $\delta t_i$  means: With  $\delta t_i$ , just as with  $\delta x^M$ , we are essentially trying to understand how the metric we are looking at/using is going to change/compare to those in different eq. classes.

As such, to get an idea of what to look at "locally" we have

$$t_i = t_i([g_{ab}])$$

$$\therefore \delta t_i = \frac{\partial t_i}{\partial g_{ab}} \delta g_{ab}$$

(Terchmüller  
Deformations)

Q: How do I know I can invert the relationship between  $t_i$  and  $g$ ?

i.e. how do I know  $\partial t_i / \partial g_{ab}$  is invertible.

I'm sure the answer is straight forward but I don't see it.

Inverting this relationship we find

$$\delta g_{ab} = \frac{\delta g_{ab}[t_i]}{\partial t_i} \delta t_i$$

This term must be added to the overall variation of  $g$ :

$$\delta g_{ab} = 2\phi g_{ab} + \nabla_a V_b + \nabla_b V_a + \delta t_i \partial_i g_{ab}$$

\* Since we know we can integrate over  $\text{Diff}_0(M)$  and  $\text{Weyl}(M)$ :

$$\Omega_{\text{Diff}_0} := \int Dg V \quad \text{and} \quad \Omega_{\text{Weyl}} := \int Dg \phi$$

where

$$\Omega_{\text{Diff}}^{[g]} = \Omega_{\text{Diff}_0}[g] \Omega_{\Gamma_g}$$

Claim:  $\Omega_{\text{Diff}_0}$  is invariant under  $\text{Weyl}$ :

Proof:

$$\Omega_{\text{Diff}_0}[e^{2\phi} \hat{g}] = \int D e^{2\phi} \mathcal{L}_V \hat{g} V = \int D e^{2\phi} \hat{g} V = \Omega[\hat{g}]$$

Gauge fixing term:  $F_{ab} = g_{ab} - g^{(f,d)}_{ab}(t) = 0$   
 $\gamma \equiv (f, \phi)$

\* We factor  $\delta g$  into (traceless) + (trace)

$$\begin{aligned} g^{ab} \delta g_{ab} &= 4\phi + 2g^{ab} \nabla_a V_b + \delta t_i g^{ab} \partial_i g_{ab} \\ &= 4\phi + 2\nabla_a V^a + \delta t_i g^{ab} \partial_i g_{ab} \\ &= 4\Delta \end{aligned}$$

Since the traceless term vanishes:

$$g^{ab} \delta g_{ab} \Leftrightarrow$$

$$\tilde{\Delta} = \phi + \frac{1}{2} (\nabla_a V^a + \delta t_i g^{ab} \partial_i g_{ab})$$

• Define  $\Delta \equiv \phi + \frac{1}{2} \nabla_a V^a$

$$\tilde{\Lambda} = \Lambda + \frac{1}{2} \delta t_i g^{ab} \partial_i g_{ab}$$

\* Plug  $\phi$  back into the expression  
 $\phi = \tilde{\Lambda} - \frac{1}{2} (\nabla_a V^a + \delta t_i g^{ab} \partial_i g_{ab})$

$$\begin{aligned} \delta g_{ab} &= 2 \left[ \tilde{\Lambda} - \frac{1}{2} (\nabla_a V^a + \delta t_i g^{cd} \partial_i g_{cd}) \right] g_{ab} \\ &\quad + \nabla_a V_b + \nabla_b V_a + \delta t_i \partial_i g_{ab} \\ &= 2 \tilde{\Lambda} g_{ab} + (\nabla_a V_b + \nabla_b V_a - g_{ab} \nabla_c V^c) \\ &\quad - \delta t_i (\partial_i g_{ab} - g_{ab} g^{cd} \partial_i g_{cd}) \end{aligned}$$

- traceless:  $2 \tilde{\Lambda} g_{ab}$
- Diffeomorphism:  $\nabla_a V_b + \nabla_b V_a - g_{ab} \nabla_c V^c \equiv (P_V)_{ab}$
- Beltrami:  $M_{ab} \equiv \partial_i g_{ab} - g_{ab} g^{cd} \partial_i g_{cd}$

\* We simply found the trace and then grouped together the terms involving  $V^a$  and  $\delta t_i$ .

$$Z_g = \int D^{\mu} g t. D_g \tilde{\Lambda} D_g (P, V) \Omega_{\text{gauge}} [g]^{-1} Z_m [g]$$

To use this Measure we must relate it to the Measure we have established over  $(V, \phi)$

• Change of coordinates:

$$(P, V, \tilde{\Lambda}) \longrightarrow (V, \phi)$$

$$D_g (P, V) D_g \tilde{\Lambda} = D_g V D_g \phi \Delta_{FP} [g]$$

The FP-Det. is simply the Jacobian

$$\Delta_{FP} [g] = \det \left( \frac{\partial (P, V, \tilde{\Lambda})}{\partial (V, \phi)} \right)$$

Formally, we write this as

$$\partial P_i V / \partial \phi = 0$$

$$\Delta_{FP}[g] = \begin{vmatrix} P_1 & 0 \\ \partial \tilde{\Lambda} / \partial V & 1 \end{vmatrix}$$

$$\begin{vmatrix} 0 & 1 \end{vmatrix}$$

$$\Delta_{FP}[g] = \det(P_1)$$

\* This is a very rough argument. A more rigorous analysis: Nakahara

### Back to West: (3.2.20)

After all that, we find that the above determinant is equivalent to Eq. 3.2.20

Since

$$\Delta_{FP} \det(P_i) = \det \left( \sqrt{-g} \frac{\delta}{\delta \zeta^\alpha} (P_i \zeta)^\alpha \right) \Big|_{\zeta^\alpha = 0}$$

$$\text{or, } \det(P_i) = \Delta_{FP} = \det \left( \frac{\delta}{\delta \zeta^\alpha} (\sqrt{-g} g^{\alpha\beta} \zeta_\beta) \right)$$

$$\Delta_{FP} = \det \left\{ \sqrt{-g} (\nabla^\alpha \zeta^\beta + \nabla^\beta \zeta^\alpha - g^{\alpha\beta} \nabla_\gamma \zeta^\gamma) \delta^2(\zeta - \zeta') \right\}$$

$$\text{Since } \delta \zeta^\alpha(\zeta) / \delta \zeta^\beta(\zeta') = \delta^\alpha_\beta \delta^2(\zeta - \zeta')$$

$$P_i: TM \rightarrow (\text{Sym}^2 TM)|_{\text{traceless}}$$

\* We have, via the inner products defined above, the adjoint

$$P_i^\dagger: (\text{Sym}^2 TM)|_{\text{traceless}} \longrightarrow TM$$

$$((P_i V)_{\alpha\beta}, t_{\alpha\beta})_2 = (V_\alpha, (P_i^\dagger t)_\alpha)_1$$

where  ${}_2$ : metric on  $\text{Sym}^2$

${}_1$ : metric on  $TM$

$$(P^\dagger t)_a = -2 \nabla^b t_{ab} \quad (\text{Adjoint Map})$$

\* Now, Recall that we have found the Action for the Ghost field and Gauge fixed term:

$$S^{gh} = -\frac{2}{\pi} \int d^2 \xi \sqrt{-g} b_{\alpha\beta} (\nabla^a c^\beta + \nabla^\beta c^a - g^{a\beta} \nabla_c c^c)$$

$$\text{or, } S^{gh} = -\frac{2}{\pi} \int d^2 \xi \sqrt{-g} b_{\alpha\beta} (P_1 c)^{\alpha\beta}$$